

The z/OS V1R12 z/OS Communications Server

Ease of use

■z/OS Communications Server is planned to enhance the z/OS Health Checker for z/OS by adding two new checks: one check for IPv4 routing and one check for IPv6 routing. The checks determine whether the total number of indirect routes in the TCP/IP stack routing table exceeds a maximum threshold (default 2000). When this threshold is exceeded, OMPROUTE and the TCP/IP stack can potentially experience high CPU consumption from routing changes.

Two new maximum threshold parameters are planned to override the default values for the total number of IPv4 and IPv6 indirect routes in a TCP/IP stack routing table before warning messages are issued.

Scalability and performance

■z/OS Communications Server AT-TLS processing will be designed to provide reduced CPU usage for encryption and decryption of application data while improving throughput for some types of workloads. This function is planned to be automatically enabled.

■z/OS Communications Server TN3270E Telnet server plans to provide access method control block (ACB) sharing for logical units (LUs) as a way to help reduce ECSA usage. Prior to z/OS V1.12 Communications Server every Telnet LU name opened its own ACB to VTAM®. You can code a new SHAREACB statement to allow multiple Telnet LUs to share a single ACB, which reduces the overall amount of ECSA (and Telnet private) storage allocated to support Telnet sessions.

Security

■z/OS Communications Server is planned to introduce trusted TCP connections, to enable sockets programs to retrieve sysplex-specific connection routing information and partner security credentials for connected sockets. Partner security credentials can be retrieved if both endpoints of a TCP connection reside in the same z/OS image, z/OS sysplex, or z/OS subplex, and the endpoints are within the same security domain. In such a topology, partner programs can use trusted connections to authenticate each other as an alternative to using an SSL/TLS connection with digital certificates for client and server authentication.

■Internet Key Exchange version 2 (IKEv2) is the latest version of the Internet Key Exchange (IKE) protocol specified by RFC 4306. IKE is used by peer nodes to perform mutual authentication and to establish and maintain security associations (SAs). In z/OS V1.12 the

Communications Server IKE daemon (IKED) is planned to be enhanced to support IKEv2, in addition to its existing IKEv1 support. The z/OS Communications Server support for IKEv2 is planned to include:

- IPv4 and IPv6 support
- A new identity type called KeyId
- Authentication using pre-shared keys or digital certificates; certificates may use RSA or elliptic curve (ECDSA) keys
- Re-keying and re-authentication of IKE SAs and child SAs
- Hash and URL specification of certificates and certificate bundles
- A new certbundle command which can create certificate bundles as specified by RFC 4306

■z/OS Communications Server is planned to introduce these enhancements to the network security services daemon (NSSD) IPsec Certificate Services:

- IKEv2 support: X.509 certificate-based signature creation and validation for IKEv2
- Elliptic Curve Digital Signature Algorithm (ECDSA) support: X.509 certificates that contain ECDSA keys may be utilized for IKEv2 digital signature creation and verification
- X.509 certificate trust chain support: The entire X.509 trust chain will be taken into consideration during IKEv1 or IKEv2 digital signature creation and verification
- Certificate Revocation List (CRL) support: CRLs may be retrieved via HTTP and consulted during IKEv1 or IKEv2 digital signature verification
- Hash and URL support: Certificates and certificate bundles specified using the Hash and URL format specified in RFC 4306 may be utilized during IKEv2 digital signature creation and verification

The z/OS Internet Key Exchange daemon (IKED) is planned to be enhanced to use these new NSSD functions when a stack is configured as a network security client.

■z/OS Communications Server is planned to introduce these enhancements to IPsec and IKE support for cryptographic currency:

- Support for the Advanced Encryption Standard (AES) algorithm in Cipher Block Chaining (CBC) using 256-bit keys, an addition to the previously existing 128-bit key support. You can use the longer key length for more-sensitive data.

- Support for the Advanced Encryption Standard (AES) algorithm in Galois Counter Mode (GCM) and in Galois Message Authentication Code (GMAC) mode. AES in GCM is intended to provide both confidentiality and data origin authentication. AES-GCM is a very efficient algorithm for high-speed packet networks. AES in GMAC mode is intended to provide data origin authentication but does not provide confidentiality. AES-GMAC, like AES-GCM, is also a very efficient algorithm for high-speed packet networks. z/OS V1.12 Communications Server is planned to support both 128-bit and 256-bit key lengths for these algorithms.

- Support for the use of Hashed Message Authentication Mode (HMAC) in conjunction with the SHA-256, SHA-384, and SHA-512 algorithms. These algorithms are intended to be used as the basis for data origin authentication and integrity verification. The new algorithms, HMAC-SHA-256-128, HMAC-SHA-384-192, and HMAC-SHA-512-256, are designed to help ensure that data is authentic and has not been modified in transit. Versions of these algorithms that are not truncated are available as Pseudo-Random Functions (PRFs). These algorithms are called PRF-HMAC-SHA-256, PRF-HMAC-SHA-384, and PRF-HMAC-SHA-512

- Support for an authentication algorithm, AES128-XCBC-96, that can help ensure data is authentic and not modified in transit.

- Support for Elliptic Curve Digital Signature Algorithm (ECDSA) authentication.

- Support for Elliptic Curve Diffie-Hellman (ECDH) key agreement

■z/OS Communications Server IPsec and IKE support is planned to leverage z/OS cryptographic modules that are designed to address the Federal Information Processing Standard (FIPS) 140-2 security requirements for cryptographic modules. FIPS 140 defines a set of security requirements for cryptographic modules to obtain higher degrees of assurance regarding the integrity of those modules. FIPS 140-2 provides four increasing, qualitative levels of security intended to cover a wide range of potential applications and environments. z/OS V1.12 Communications Server support is planned to be configurable such that it will only utilize underlying security modules (System SSL and ICSF's PKCS #11 capabilities) that are operating in FIPS 140 mode. System SSL and ICSF's PKCS #11 capabilities are designed to address the requirements for FIPS 140-2 level 1.

■RFC 4301 compliance for IPsec filter rules is planned to become mandatory. RFC 4301 "Security Architecture for the Internet Protocol" specifies the base architecture for IPsec-compliant systems, including restrictions on the routing of fragmented packets. Compliance enforcement may require minor changes to IP filters for IP traffic that is routed through z/OS. The Configuration Assistant will be designed to identify any non-compliant IP filters and policy agent will not install an IPsec policy that contains any non-compliant IP filters.

Availability

■z/OS Communications Server plans to introduce sysplex distributor support for a hot-standby server through the use of a new distribution method, HotStandby. You configure a preferred server and one or more hot-standby servers. The preferred server with an active listener receives all new incoming connection requests, and the hot-standby servers act as backup servers should the designated preferred server become unavailable. The hot-standby servers can be ranked to control which hot-standby server becomes the active server. You can also control whether the sysplex distributor automatically switches back to using the preferred server if it again becomes available, and whether the distributor automatically switches servers if the active target is not healthy.

■z/OS Communications Server sysplex problem detection and recovery is planned to be enhanced to detect when the TCP/IP stack has abended five times in less than a minute. Existing sysplex recovery logic is applied when this problem is detected.

Optimization and management capabilities

■In z/OS V1.12, z/OS Communications Server is planned to use new TCP/IP callable NMI requests to provide TCP/IP stack network interface information and network interface and global statistics. Network management applications can use the requested output to monitor interface status and TCP/IP stack activity. z/OS V1.12 Communications Server is planned to provide the following new requests:

- GetGlobalStats - Provides TCP/IP stack global counters for IP, ICMP, TCP, and UDP processing
- GetIfs - Provides TCP/IP network interface attributes and IP addresses
- GetIfStats - Provides TCP/IP network interface counters
- GetIfStatsExtended - Provides data link control (DLC) network interface counters

■z/OS Communications Server plans to provide enhancements to improve the management of the CSSMTP application by adding the following new SMF 119 record subtypes:

- 048 - CSSMTP Configuration data records
- 049 - CSSMTP Target server connection records
- 050 - CSSMTP Mail records
- 051 - CSSMTP Spool records
- 052 - CSSMTP Statistics records

It is intended that applications that want to process the new SMF 119 subtypes can obtain them from a traditional MVS SMF exit routine or in real time from the z/OS Communications Server Network Management Interface (NMI) for SMF, SYSTCPSM.

CSSMTP issues the SIOCSAPPLDATA ioctl to add application data (appldata) to the TCP connections used to connect to target mail servers. You can see the application data (appldata) displayed in the Netstat All/-A, AllConn/-a, and Conn/-c reports.

■z/OS Communications Server plans to introduce sysplex event notification through new SMF 119 event records (subtypes 32 - 37) that describe the following events:

- DVIPA status change (subtype 32)
- DVIPA removed (subtype 33)
- DVIPA target added (subtype 34)
- DVIPA target removed (subtype 35)
- DVIPA target server started (subtype 36)
- DVIPA target server ended (subtype 37)

The new SMF 119 event records are planned to be written to the MVS SMF data sets, and can also be obtained from the real-time TCP/IP network monitoring NMI (SYSTCPSM).

■z/OS Communications Server planned improvements include:

- The ability to learn indirect prefix routes from IPv6 Router Advertisement messages
- The ability to associate preference values with all routes that are learned from IPv6 Router Advertisement messages

Use of these functions is expected to reduce the number of IPv6 static routes that must be defined and the ability to route around network failures when not using OMPROUTE to install routes learned via a dynamic routing protocol, such as OSPF.

Networking

■z/OS Communications Server V1.12 is planned to provide notification to the operator console when a Domain Name System (DNS) name server does not respond to a certain percentage of resolver queries sent to the name server during a sliding five-minute interval. In addition to the notification, statistics regarding the number of queries attempted and the number of queries that received no response are displayed for each currently unresponsive name server at five-minute intervals. This can alert you to a possible problem with your

DNS name server configuration that may be adversely affecting applications on your z/OS system. The default value for the TCPIP.DATA RESOLVERTIMEOUT configuration statement, which controls the timeout value for UDP requests sent to a name server, is planned to be modified to be five seconds instead of 30 seconds.

■z/OS Communications Server plans to extend the VARY TCPIP,,DROP command to allow the dropping of all established TCP connections for servers that match the specified filter parameters. When issued, each server that is found to match the specified filter parameters will have all its established TCP connections dropped. You can filter by port, jobname, or server ASID. This function is expected to make it easier to move workload from one application instance to another application instance.

■z/OS Communications Server is planned to provide the option of keeping a TCP/IP stack isolated from the sysplex; you can use a new configuration parameter to prevent a stack from automatically joining the sysplex group at startup. You can have the stack join the sysplex group at a later time by issuing the VARY TCPIP,,SYSPLEX,JOINGROUP command.

■z/OS Communications Server is planned to enhance the performance of fast local sockets for TCP connections. This function is planned to be automatically enabled.

■z/OS Communications Server provides local and path MTU discovery to learn the correct MTU size for Enterprise Extender (EE) connections. The MTU size is used to modify the link size for EE connections. In z/OS V1.12, the link size is planned to be updated at the RTP pipe endpoints in addition to the EE endpoints when the MTU size changes.

■z/OS Communications Server packet trace filtering is planned to be enhanced to support:

- Including the next hop IP address on the trace output. This can be obtained from the fully formatted packet trace using IPCS. The next hop IP address is also available to applications that consume the real-time packet trace through the real-time TCP/IP networking monitoring API.
- Making packet trace filtering available to encapsulated packets that are used in VIPAROUTE traffic.

■z/OS Communications Server is planned to provide the option to check the health of an Enterprise Extender (EE) connection during the activation of the connection. The health of active connections can also be verified.

■z/OS Communications Server is planned to reduce CPU utilization for TCP/IP Callable Network Management Interface (NMI), EZBNMIFR, GetConnectionDetail. All the filters that are specified for the request must contain the complete identification (4-tuple) of established TCP connections. The 4-tuple of a TCP connection consists of the local IP address, local port, remote IP address, and remote port for the connection.

■The z/OS Communications Server Netstat function is planned to provide support for verifying that message catalogs being used are at the correct level when they are opened. This function is intended to prevent Netstat from abending or not functioning correctly when the message catalog is out of synch with the Netstat program.

■z/OS Communications Server enhances TCP/IP data tracing (DATTRACE) to provide two new trace records:

- A Start record with State field "API Data Flow Starts" that indicates the first data sent or received by the application for the associated TCP or UDP socket
- An End record with State field "API Data Flow Ends" that indicates the socket has been closed

■z/OS Communications Server is designed to support RFC3484 by providing a configurable policy table for default address selection for IPv6. The source address selection algorithm and destination address selection algorithm are planned to be enhanced to support additional address selection rules in conjunction with the configured or default policy table. For example, you might choose to prefer IPv4 communication over IPv6 by providing a custom policy table for default address selection.

■z/OS Communications Server is also planned to support RFC5014 by providing IPv6 socket API for source address selection. Applications can indicate they prefer temporary IPv6 addresses over public IPv6 addresses or public IPv6 addresses over temporary IPv6 addresses.

■Additionally, z/OS Communications Server is planned to enhance the SRCIP configuration to allow an administrator to indicate that the TCP/IP stack should prefer public IPv6 addresses over temporary IPv6 addresses. This will allow you to override the preferences specified by an application using the IPv6 socket API for source address selection.

■z/OS Communications Server is planned to allow the system resolver to send requests to Domain Name System (DNS) name servers using IPv6 communication. This function is planned to allow you to use the existing NSINTERADDR and NAMESERVER resolver configuration statements in the TCPIP.DATA dataset to define the IPv6 address of the name server.

■z/OS Communications Server allows the coding of MULTIPATH in the TCP/IP profile that enables multipath support for IP packets. You might want this behavior for TCP connections but not for Enterprise Extender (EE) connections. In z/OS Communications Server V1.12, the multipath function is planned to be disabled by default for EE connections regardless of the value specified in the TCP/IP profile. You can use the VTAM start option MULTIPATH to control the multipath function for EE.

■z/OS Communications Server plans to enhance the digital certificate access server (DCAS) to allow modification of the debug level without restarting the application.

Statement of general direction

■IBM plans to pursue an evaluation to the Federal Information Processing Standard (FIPS) 140-2 using National Institute of Standards and Technology's (NIST) Cryptographic Module Validation Program (CMVP) for the PKCS #11 capabilities of the Integrated Cryptographic Service Facility (ICSF) component of the Cryptographic Services element of z/OS. The scope of this evaluation will include algorithms provided by the CP Assist for Cryptographic Functions (CPACF) that are utilized by ICSF. This is intended to help satisfy the need for FIPS 140-2 validated cryptographic functions when using z/OS Communications Server capabilities such as the IPsec protocol.

NssD Overview.

This topic contains a description of IP filtering, IPsec-protected traffic, and preparing and configuring a z/OS® system for IP security. Various business configurations are explained, including host-to-host, host-to-gateway, gateway-to-gateway, and gateway-to-hos

A network security services (NSS) server provides network security services for one or more security disciplines. Supported disciplines are IPsec, which is a set of services that supports IPsec and IKE processing, and XMLAppliance, which is a set of services for XML appliances. For the IPsec discipline, these services include the IPsec certificate service and the IPsec remote management service. For the XMLAppliance discipline, the NSS server supports the XMLAppliance SAF access service, the XMLAppliance certificate service, and the XMLAppliance private key service. For information about configuring the IKE daemon to act as an NSS IPsec client on behalf of a TCP/IP stack